

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence
NAME: R. RUPA SREE

COURSE CODE: CSA1714
REGISTER NUMBER: 192472122

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5
Marks:50

Duration : 1 Hour
Total

Annexure A

Design Task

Code of Conduct:

I R. Rupa Sree (192472122) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
R. RUPA SREE

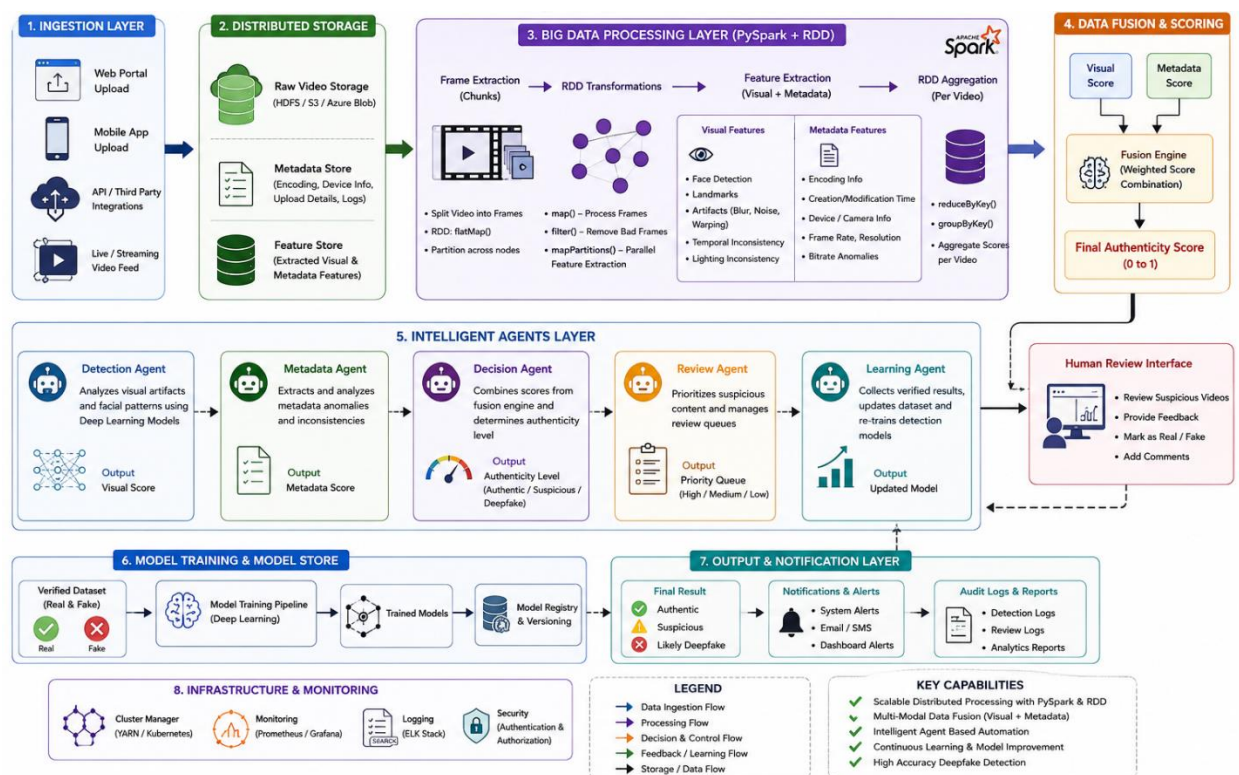
Design Task

Deepfake Detection and Face Authenticity Verification Platform

Design a distributed AI platform using PySpark and RDD operations to analyse large volumes of uploaded video content for deepfake detection and face authenticity verification. Describe the system architecture, the data fusion techniques used to combine visual artifacts and metadata analysis, and how intelligent agents coordinate to flag suspicious content, prioritize review queues, and continuously retrain detection models.

ANS:

1. System Architecture



2. PySpark and RDD Operations

PySpark distributes large video collections across multiple worker nodes.

RDD operations used:

- **map()** – extract frames and video features.
- **filter()** – remove invalid or low-quality videos.
- **flatMap()** – generate multiple frames from each video.
- **reduceByKey()** – combine detection scores.
- **groupByKey()** – group results belonging to the same video.

This allows thousands of videos to be processed in parallel.

3. Data Fusion

Two types of information are combined:

Visual Analysis

- Face landmarks
- Blurring and warping artifacts
- Lighting inconsistencies
- Facial expression inconsistencies
- Frame-to-frame changes

Metadata Analysis

- Video format
- Encoding information
- Creation/modification details
- Frame rate and resolution
- Suspicious metadata changes

The scores from both sources are combined to produce a **final authenticity score**.

Example:

$$FinalScore = 0.7(VisualScore) + 0.3(MetadataScore)$$

A high score indicates a higher probability of deepfake content.

4. Intelligent Agent Coordination

Different agents perform different tasks:

- **Detection Agent:** analyzes faces and visual artifacts.
- **Metadata Agent:** checks metadata anomalies.
- **Decision Agent:** combines scores and determines authenticity.
- **Review Agent:** prioritizes highly suspicious videos for human review.
- **Learning Agent:** collects verified results and retrains the detection model.

5. Continuous Learning

New Videos



Detection



Suspicious Videos



Human Verification



Verified Data



Model Retraining



Improved Detection Model

6. Final Outcome

The platform provides:

- **Authentic**
- **Suspicious**
- **Likely Deepfake**

The distributed PySpark architecture enables **scalable processing**, while data fusion improves detection reliability. Intelligent agents automate detection, review prioritization, and continuous model improvement.

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	Data Flow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete
Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations/actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration
Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed
Data Flow & Processing Pipeline	Clear, logical, and well-defined data flow	Mostly clear	Some gaps	Poorly defined
Innovation & Creativity	Highly innovative, realistic, and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation & Documentation	Very clear, neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poor presentation